

Matrix Product States

$$|\psi\rangle = \sum_{s_1 \dots s_L} \underbrace{A^{[1]s_1} \dots A^{[L]s_L}}_{\text{Product of } L \text{ matrices, one per site}} |s_1\rangle \otimes \dots \otimes |s_L\rangle$$

$s_i = 1, \dots, d$
 $s_i = \uparrow, \downarrow$

$A^{[1]s_1}$: Row vector (r_1 entries)

$A^{[i]s_i}$: Matrix of dim. $r_{i-1} \times r_i$

$A^{[L]s_L}$: Column vector (r_{L-1} entries)

$$\begin{pmatrix} & & & \\ & & & \\ 1 \times r_1 & & & \end{pmatrix} \begin{pmatrix} & & & \\ & & & \\ & & & \\ r_1 \times r_2 & & & \end{pmatrix} \dots \begin{pmatrix} & & & \\ & & & \\ & & & \\ r_{L-2} \times r_{L-1} & & & \end{pmatrix} \begin{pmatrix} & & & \\ & & & \\ & & & \\ r_{L-1} \times 1 & & & \end{pmatrix} = \text{number}$$

Maximum value of $r_i = \min(d^i, d^{L-i})$

Max dim. of MPS:

$$\underbrace{(1 \times d)}_{A^{[1]}} \underbrace{(d \times d^2)}_{A^{[2]}} \dots \underbrace{(d^{L/2-1} \times d^{L/2})}_{A^{[L/2]}} \underbrace{(d^{L/2} \times d^{L/2 \times 1})}_{A^{[L/2+1]}} \dots \underbrace{(d \times 1)}_{A^{[L]}}$$

Product state:

$$|4\rangle = |s_1^0\rangle \otimes |s_2^0\rangle \otimes \dots \otimes |s_L^0\rangle$$

$$|\uparrow\rangle \otimes |\uparrow\rangle \otimes \dots \otimes |\uparrow\rangle$$

$$= |s_1^0 \dots s_i^0\rangle \otimes |s_{i+1}^0 \dots s_L^0\rangle \quad r_i = 1 \quad \forall i$$

$\Rightarrow$ All matrices must be 1×1 .

$$A^{[i]s_i} = \delta_{s_i, s_i^0} \quad \left(= \delta_{s_i, \uparrow} \text{ for ferromagnet } |\uparrow\uparrow\dots\uparrow\rangle \right)$$

GHZ state:

$$|4\rangle = \frac{1}{\sqrt{2}} (|000\rangle + |111\rangle) \quad L=3$$

$$C_{000} = C_{111} = 1/\sqrt{2}$$

Step 1: 

* Reshape $C_{s_1 s_2 s_3}$ into $d \times d^2$ matrix:

$$C = \begin{pmatrix} C_{000} & C_{001} & C_{010} & C_{011} \\ C_{100} & C_{101} & C_{110} & C_{111} \end{pmatrix} = \begin{pmatrix} 1/\sqrt{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/\sqrt{2} \end{pmatrix}$$

$$C = U \Sigma V^T = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \underbrace{\begin{pmatrix} 1/\sqrt{2} & 0 \\ 0 & 1/\sqrt{2} \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}}_{\hat{C}} = \begin{pmatrix} 1/\sqrt{2} & 0 & 0 & 0 \\ 0 & 0 & 0 & 1/\sqrt{2} \end{pmatrix}$$

$$A^{[1]0} = \begin{pmatrix} 1 & 0 \end{pmatrix}$$

$$A^{[1]1} = \begin{pmatrix} 0 & 1 \end{pmatrix}$$

Step 2: ~~0~~ | ~~0~~ | ~~0~~

* Reshape $\hat{C}$ into $(r_1, d) \times d^1 = (2, 2) \times 2 = 4 \times 2$

$$\tilde{C} = \begin{pmatrix} 1/\sqrt{2} & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 1/\sqrt{2} \end{pmatrix} = \underbrace{\begin{pmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 0 \\ 0 & 1 \end{pmatrix}}_U \underbrace{\begin{pmatrix} 1/\sqrt{2} & 0 \\ 0 & 1/\sqrt{2} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}}_{\tilde{C}} = \begin{pmatrix} 1/\sqrt{2} & 0 \\ 0 & 1/\sqrt{2} \end{pmatrix}$$

$$A^{[2]0} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$A^{[2]1} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

Step 3

$$A^{[3]0} = \begin{pmatrix} 1/\sqrt{2} \\ 0 \end{pmatrix}$$

$$A^{[3]1} = \begin{pmatrix} 0 \\ 1/\sqrt{2} \end{pmatrix}$$

$$C_{s_1 s_2 s_3} = A^{[1]s_1} A^{[2]s_2} A^{[3]s_3}$$

$$C_{000} = \begin{pmatrix} 1 & 0 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 1/\sqrt{2} \\ 0 \end{pmatrix} = \underline{\underline{1/\sqrt{2}}}$$

$$|4\rangle = \sum_{s_1 s_2 s_3} C_{s_1 s_2 s_3} |s_1 s_2 s_3\rangle = \sum_{s_1 s_2 s_3} A^{[1]s_1} A^{[2]s_2} A^{[3]s_3} |s_1 s_2 s_3\rangle$$

$$|4\rangle = \frac{1}{\sqrt{2}} (|100 \dots 0\rangle + |111 \dots 1\rangle) \quad L \text{ sites}$$

$$10 \dots 0 \overset{A}{\rangle} \otimes 10 \dots 0 \overset{B}{\rangle} + |1 \dots 1\rangle \otimes |1 \dots 1\rangle$$

$$r_i = 2 \quad \forall i$$

Since $r_i = 2$: - we have 2 "channels"

* build $|100 \dots 0\rangle$

* build $|111 \dots 1\rangle$

- $A^{[i]s_i}$ is a 2×2 matrix.

Start with $|100 \dots 0\rangle$:

$$A^{[1]0} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$A^{[2]0} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$A^{[1]0} A^{[2]0} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$A^{[i]0} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}$$

$$A^{[L]0} = \begin{pmatrix} 1/\sqrt{2} \\ 0 \end{pmatrix}$$

Similarly for $|11\dots 1\rangle$:

$$A^{[1]1} = \begin{pmatrix} 0 & 1 \end{pmatrix}$$

$$A^{[i]1} = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

$$c_{1\dots 1} = 1/\sqrt{2}$$

$$A^{[2]1} = \begin{pmatrix} 0 \\ 1/\sqrt{2} \end{pmatrix}$$

$$A^{[i]0} A^{[i+1]1} = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} = 0$$

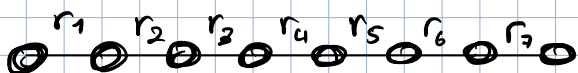
Generally, an exact MPS would need d^2 parameters, same as C .

However, when entanglement is small, $|\psi\rangle$ can be well approximated by truncation, i.e. restrict the rank of each matrix / bond of the MPS.

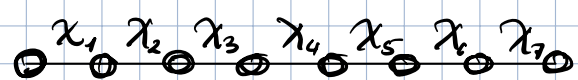
How to truncate:

- * At each bond (for each SVD), keep up to χ Schmidt values / singular values.

Before:



After:



$$\chi_i = \min(r_i, \chi)$$

By Eckhart-Young: this is the best rank- χ approximation to each matrix.

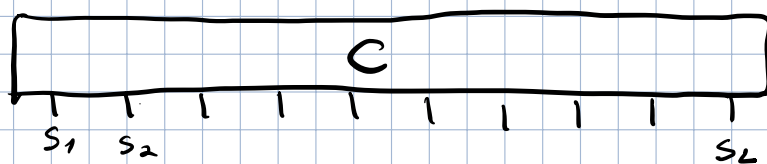
Truncated state: $|\psi_\chi\rangle$ is variational

$$\frac{\langle \psi_\chi | H | \psi_\chi \rangle}{\langle \psi_\chi | \psi_\chi \rangle} \geq E_0$$

$|\psi_\chi\rangle$ has at most $L \cdot d \cdot \chi^2$ parameters (as opposed to d^L for $|\psi\rangle$)

Diagrammatic representation of MPS

$$|\psi\rangle = \sum_{s_1 \dots s_L} C_{s_1 \dots s_L} |s_1 \dots s_L\rangle$$



As an MPS:



Generally:

A vector: $a - \boxed{v} = \vec{v}_a$

A matrix: $a - \boxed{M} - b = M_{ab}$

$$(Mv)_a = \sum_b M_{ab} v_b = a - \boxed{M} - \boxed{v}$$

MPS

$$|\psi\rangle = \sum_{a_1 \dots a_{L-1}} \sum_{s_1 \dots s_L} A_{a_1}^{[1]s_1} A_{a_1 a_2}^{[2]s_2} \dots A_{a_{L-1}}^{[L]s_L} |s_1 \dots s_L\rangle$$

$$A_{a_{i-1} a_i}^{[i]s_i} = a_{i-1} - \boxed{} - a_i$$

s_i

$$|\psi\rangle = \sum_{s_1 \dots s_L} \boxed{}_{s_1} - \boxed{} - \boxed{} - \dots - \boxed{} - \boxed{}_{s_L} |s_1 \dots s_L\rangle$$

* Physical indices are vertical legs

* MPS bond indices are horizontal legs

Gauge freedom / Canonical forms

The MPS is not unique.

We can insert $X X^{-1}$ on any bond.

$$A^{[i]s_i} \rightarrow A^{[i]s_i} X$$

$$A^{[i+1]s_{i+1}} \rightarrow X^{-1} A^{[i+1]s_{i+1}}$$

$\Rightarrow$ χ^2 numbers on each bond to be fixed by a chosen gauge.

We use this freedom to define some convenient MPS known as "canonical forms"

$$\text{Left-canonical: } \sum_{s_i} A^{[i]s_i \dagger} A^{[i]s_i} = \mathbb{1}$$

$$(U^\dagger U = \mathbb{1})$$

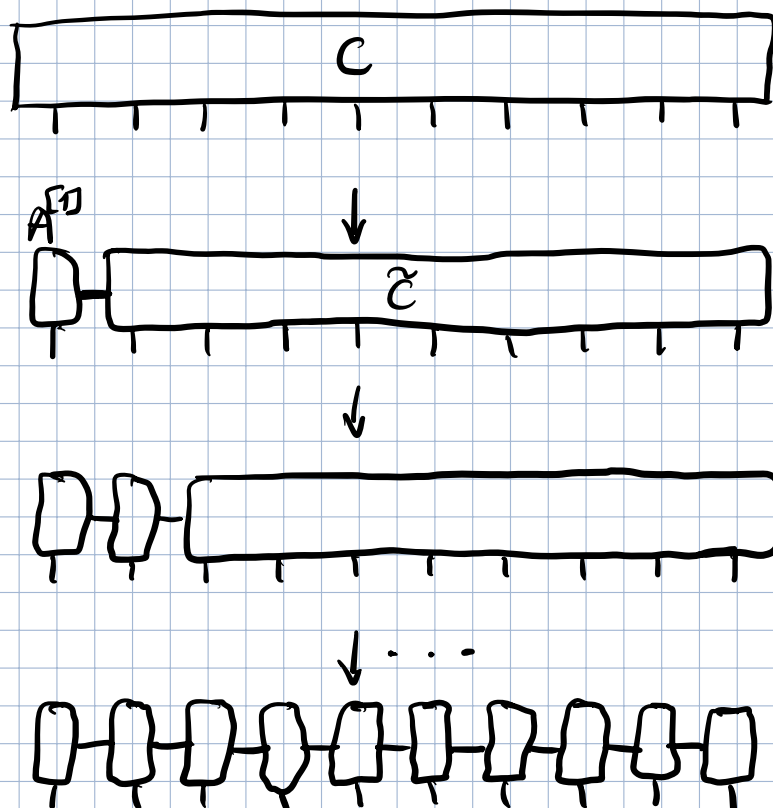
$$\text{Right-canonical: } \sum_{s_i} B^{[i]s_i} B^{[i]s_i \dagger} = \mathbb{1}$$

$$(V^\dagger V = \mathbb{1})$$

B is extracted from $V^\dagger$.

$$C = U \Sigma V^\dagger$$

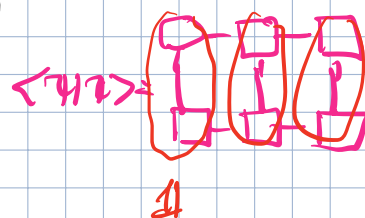
Left-canonical:



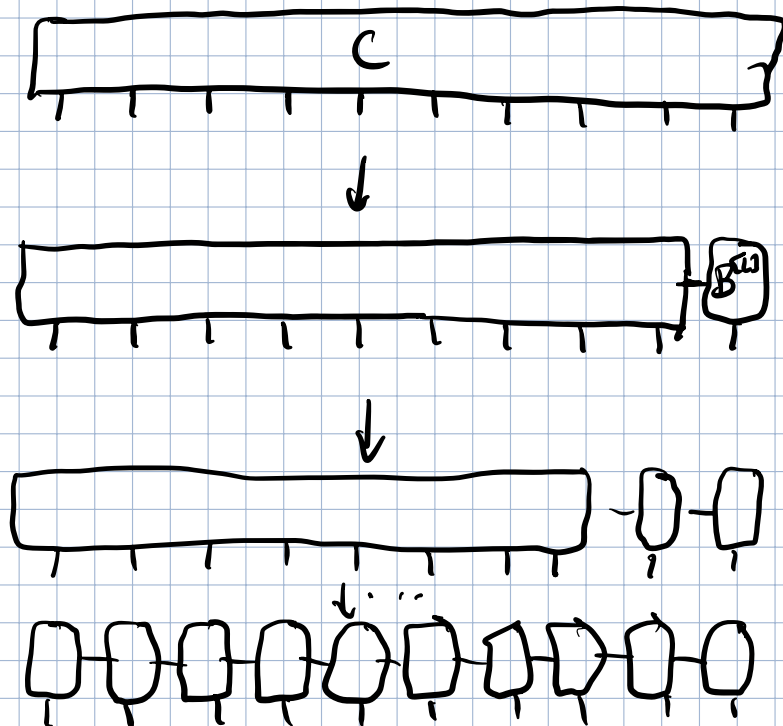
$$C = U \Sigma V^T$$

$\uparrow$
 $\sim A$

$$A^T A = \mathbb{1}^n$$



Right-canonical:



$$C = U \Sigma V^T$$

$\uparrow$
 $\sim B$

Mixed canonical form:

- * Left-canonical to the left of bond n
- * Right-canonical to the right of bond n

